System Administration and Maintenance

Presented by:
Lieutenant Commander (IT) RMJP Nayanajith

Year 2 Semester 2			
Course Code	ITIC2212		
Course Name	System Administration and Maintenance		
Credit Value	2		
Core/Optional	Core		
Hourly Breakdown	Theory	Practical	Independent Learning
	30	30	90
Course Aim	To impart the fundamental skills and best practices necessary for managing modern computer network systems.		
Intended Learning Outcomes	After the completion of this module, the student will be able to, **LO1** Describe Concepts on computer systems **LO2** Describe fundamental concepts on computer networks **LO3** Demonstrate an understanding of the components of a Computer Network **LO4** Implement a computer network for the existing requirement **LO5** Identify network configurations		
Course Contents	**Introduction to Computer Networks** Introduction to computer network, Features, Architecture, Network types, Topologies, Transmission modes		**LO1**
	Communication concepts Internet protocol suite, TCP/IP, DNS, FTP, SMTP		**LO2**
	Hardware components on computer networks The hardware that makes up the physical part of the network, and configuring the files or databases, hosts, routers, and network configuration servers.		**LO3**
	Implementation Computer network planning, design and implementation.		**LO3** **LO4**
	Software configurations on computer Networks The process of assigning network settings, policies, flows, and controls.		**LO3** **LO4** **LO5**
Teaching/ Learning Methods	Lectures, demonstrations, practical sessions, learner-centred learning approaches, discussion, and group interactions.		
Assessment Strategy	Continuous and Semester end assessment		

Continuous Assessments 40%	**Laboratory/ Practical Sessions (20%)**	• Lab 1: Network classification (5 %) • Lab 2: How to build a simple network (5 %) • Lab 3: Software configuration (5 %) • Lab 4: Linux based net configurations (5 %)	**LO 2 LO 3 LO 4 LO 5**
	Assignments/ Tutorials (10%)	• Assignment 1 - Computer Networks (2.5%) • Assignment 2 - Communication concepts (2.5%) • Assignment 3 - Hardware components on computer networks (2.5%) • Assignment 4 - Software configurations on computer Networks (2.5%)	**LO1-LO5**
	Field visits/ Onsite training (5%)	• Field Visit 01 (5%)	**LO2-LO5**
	Mid-Term (5%)	• Mid semester examination – 2 hours structured essay question paper (5%)	**LO1-LO3**
Final Assessment 60 %		• Structured Essay 3h	**LO1-LO5**
References		1. B. Forouzan A., Data Communications and Networking. McGraw-Hill Education, 5th Edition, 2013. 2. Frisch, A. (2002). Essential System Administration: Tools and Techniques for Linux and Unix Administration. US: O'Reilly Media	

01. Introduction to Computer Networks

- Introduction to Computer Networks
- Features
- Network Architecture
- Network Types
- Network Topologies
- Transmission Modes

Learning Outcomes

After this lesson, students will be able to **Describe Concepts on computer Network systems**

- Define a computer network.
- Explain the importance of networking.
- Describe the features of computer networks.
- Understand network architecture.
- Identify different network types.
- Compare various network topologies.
- Explain transmission modes.

What is a Computer Network

A **computer network** is a collection of interconnected computers and devices that communicate with each other to share resources, data, and services.

* **Examples:**
  - Internet
  - Office network
  - University campus network
  - Home Wi-Fi network

* **Key Components:**
  - Computers
  - Servers
  - Switches
  - Routers
  - Network cables/Wireless Access Points

Basic Requirements to Create a Network

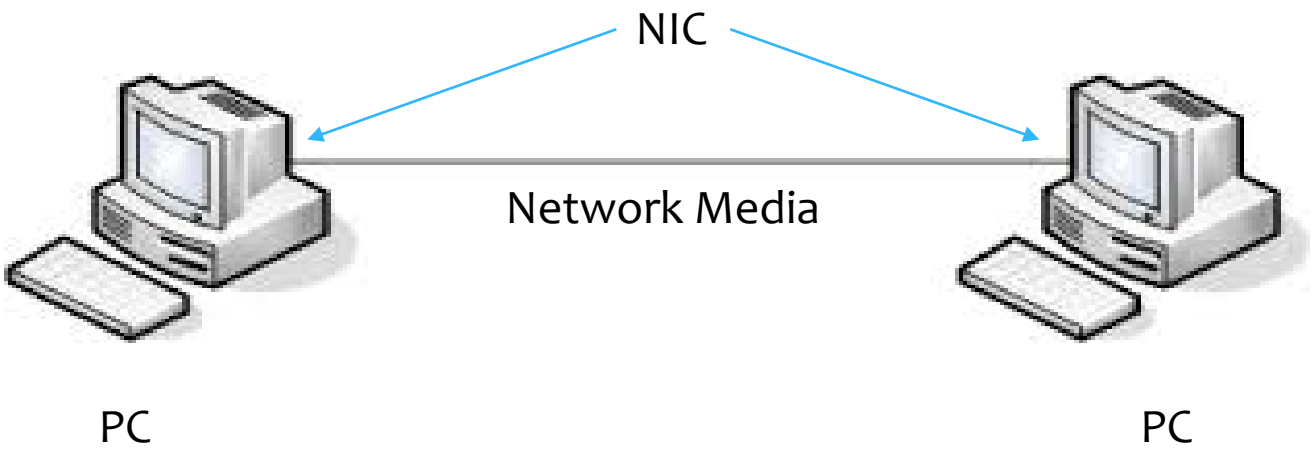
1. Two Personal Computer (PCs).

2. NIC (Network Interface Card).

3. Network Media.

4. Protocol. (Set of rules and regulations / Standard).

5. Nos (Network Operating System)



Why Do We Need Computer Networks

* **Benefits**
  - Resource sharing
  - File sharing
  - Internet access
  - Communication (Email, Chat, Video calls)
  - Centralized data management
  - Remote access
  - Cloud computing

* **Example :**

Students in a computer lab access the same printer and Internet connection.

Features of Computer Networks

- **Major Features**
- Resource Sharing
- High Reliability
- Scalability
- Fast Communication
- Security
- Cost Effectiveness
- Centralized Administration

Feature	**Description**
Resource Sharing	Share files, printers, software
Reliability	Backup paths and redundancy
Scalability	Easy to add devices
Security	Authentication and encryption

Network Architecture

Network architecture defines how devices communicate and how services are organized.

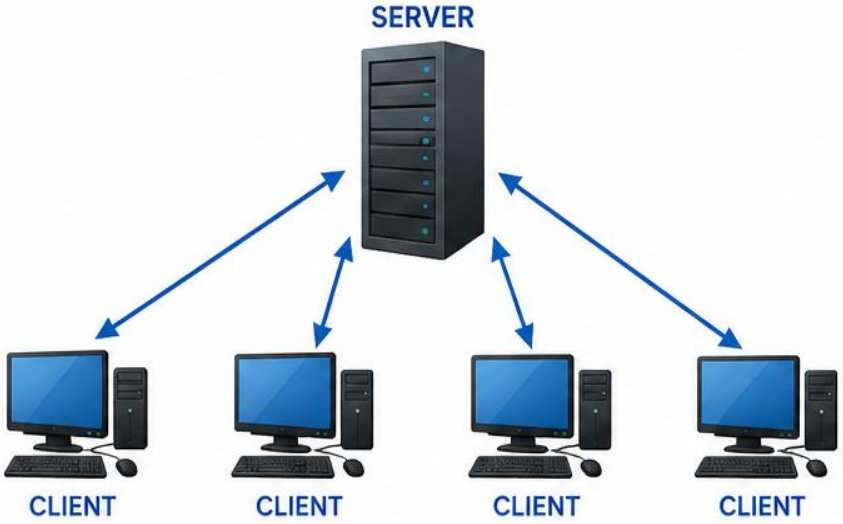
There are 02 Types

1. **Client-Server Architecture**
   - Dedicated server
   - Centralized management
   - Better security
   - Suitable for organizations

2. **Peer-to-Peer (P2P)**
   - No dedicated server
   - Each computer acts as both client and server
   - Easy setup
   - Suitable for small networks

CLIENT–SERVER NETWORK

All clients request services from a dedicated server.



FEATURES

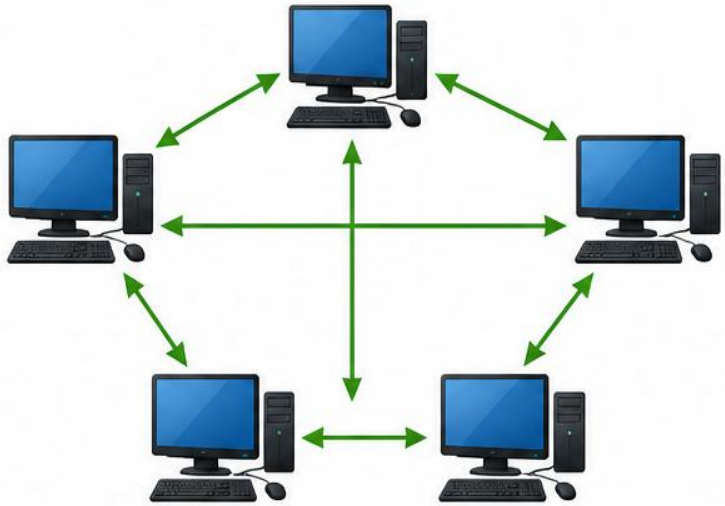
- Dedicated server provides resources and services.
- Centralized management and security.
- Easy backup and data management.
- Higher cost.
- Suitable for medium to large networks.

EXAMPLES

Print Server File Server Web Server

PEER-TO-PEER NETWORK

All computers (peers) are equal and can act as both client and server.



FEATURES

- No dedicated server; every peer shares resources.
- Decentralized management.
- Low cost and easy to setup.
- Less secure than client–server.
- Suitable for small networks.

EXAMPLES

File Sharing Printer Sharing Media Sharing

Client-Server vs Peer-to-Peer

Client-Server	Peer-to-Peer
Dedicated server	No dedicated server
High security	Lower security
Easy centralized management	Difficult management
Expensive	Low cost
Large organizations	Home or small office

Types of Networks

- **PAN (Personal Area Network)**
  - Distance – Computers can be connected within **1 – 10 meters**
  - Bluetooth
  - Mobile hotspot

- **LAN (Local Area Network)**
  - Distance – Computers can be connected within **10 - 100 meters**
  - Home
  - School
  - Office

- **MAN (Metropolitan Area Network)**
  - Distance – Computers can be connected More Than **100 meters – Lesser Than 5KM**
  - City-wide network

- **WAN (Wide Area Network)**
  - Distance – Computers can be connected **More Than 5KM range**.
  - Country or worldwide
  - Internet

Types of Networks

Type	Coverage	Example
• **PAN**	Personal	Bluetooth
• **LAN**	Building	School lab
• **MAN**	City	City network
• **WAN**	Worldwide	Internet

Network Topologies

What is a Topology

A **network topology** refers to the physical or logical arrangement of devices in a network.

Common topologies:

- Bus
- Star
- Ring
- Mesh
- Tree
- Hybrid

Bus Topology

* **Characteristics**
  - Single backbone cable
  - All devices connected to one cable

* **Advantages**
  - Low cost
  - Easy installation

* **Disadvantages**
  - Backbone failure affects entire network
  - Difficult troubleshooting

Star Topology

* **Characteristics**
  - All devices connected to a central switch/hub. Most common topology today

* **Advantages**
  - Easy maintenance
  - Easy expansion
  - Failure of one computer doesn't affect others

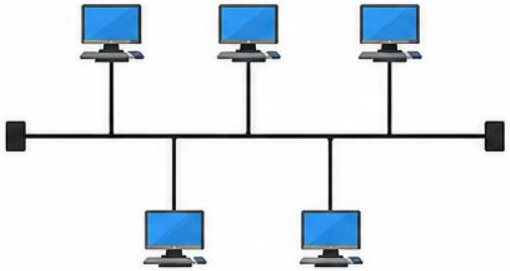
* **Disadvantages**
  - Central device failure stops the network

Ring, Mesh, Tree & Hybrid

* **Ring**
  - Devices connected in a circle
  - Data travels around the ring
* **Mesh**
  - Every device connected to every other device
  - Very reliable
  - Expensive
* **Tree**
  - Hierarchical structure
  - Used in organizations
* **Hybrid**
  - Combination of two or more topologies

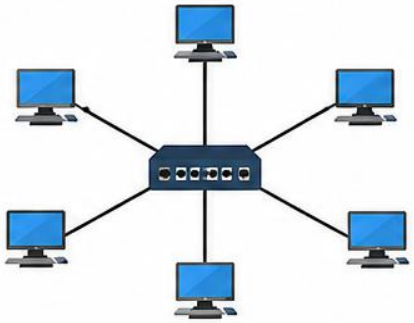
Network Topologies

1. Bus Topology



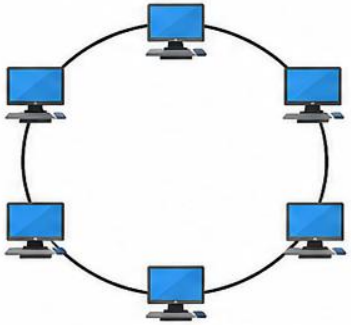
- All devices are connected to a single backbone cable.
- Easy to install and low cost.
- Backbone failure stops the entire network.

2. Star Topology



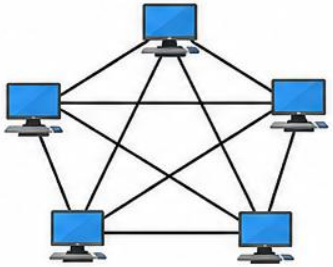
- All devices are connected to a central hub/switch.
- Easy to install, manage and expand.
- Failure of one device doesn't affect others.
- Central device failure stops the network.

3. Ring Topology



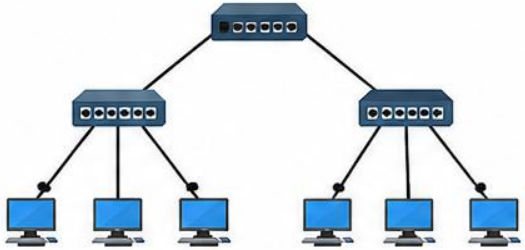
- Each device is connected to two other devices.
- Data travels in one direction around the ring.
- Failure of one device can affect the entire network.

4. Mesh Topology



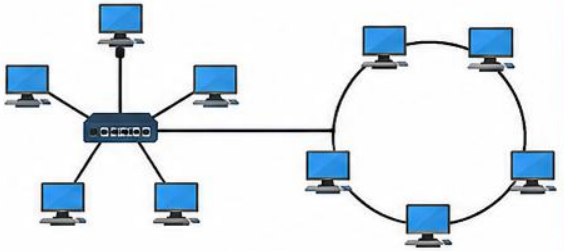
- Every device is connected to every other device.
- Very reliable and fault tolerant.
- High cost and complex installation.

5. Tree Topology



- Hierarchical structure with root and branches.
- Easy to manage and scalable.
- Backbone failure affects entire network.

6. Hybrid Topology



- Combination of two or more topologies.
- Flexible and scalable.
- Complex design and high cost.

Comparison of Topologies

Topology	Cost	Reliability	Easy Expansion
• **Bus**	Low	Low	Moderate
• **Star**	Medium	High	High
• **Ring**	Medium	Medium	Low
• **Mesh**	Very High	Very High	Difficult
• **Tree**	Medium	High	High

Transmission Modes

Transmission mode defines the direction in which data flows between devices.

Three modes:

- Simplex
- Half Duplex
- Full Duplex

Simplex Transmission

Data flows in **one direction only.**

Examples:

- Keyboard → Computer
- Television broadcasting
- Radio

Sender ----------> Receiver

Half Duplex Transmission

Data flows in **both directions but one direction at a time.**

Examples:

- Walkie-talkies
- Two-way radio

A <----------> B

(One direction at a time)

Full Duplex Transmission

Data flows in **both directions simultaneously.**

Examples

- Telephone
- Video calls
- Mobile communication

A <==========> B

(Both directions simultaneously)

Comparison of Transmission Modes

Mode	Direction	Example
Simplex	One-way	Television
Half Duplex	Two-way (one at a time)	Walkie-talkie
Full Duplex	Two-way simultaneously	Telephone

THANK YOU